# Flipping Contractions in the Kähler MMP

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**Big picture**

In the projective MMP the existence of flips is the hard analytic core, settled by finite generation of the log canonical algebra (BCHM). Passing to the compact Kähler / Fujiki-class setting *redistributes* this difficulty. On the one hand, the Kähler notion of a flip does not require the contraction to be projective, so once a flipping *contraction* $f\colon X \to Z$ is available, producing the flip is comparatively soft — a local finite-generation / Proj statement. On the other hand, constructing the flipping contraction itself — extending a small $(K_X + B)$-negative extremal ray to an honest bimeromorphic morphism onto a Kähler base — is now where the real work lies, because a supporting class of the ray is only a transcendental $(1,1)$-class, not a divisor.

This note develops that story for compact Kähler 3-folds and contrasts the **two methods** available: the analytic-positivity approach of **Höring–Peternell** [?] (terminal case), which contracts the flipping curve directly by producing an ample conormal bundle and invoking Grauert's theorem; and the **Das–Hacon** [?, ?] approach for klt/dlt log pairs, which bootstraps the full log MMP on a dlt modification. Section 1 fixes the flip formalism, Section 2 sets up the analytic framework and the common "flip = finite generation" engine, Sections 3–4 give the two methods, and Section 5 compares them.

## 1 Flipping contractions and the need for flips

We work throughout with normal compact analytic varieties in Fujiki's class $\mathcal{C}$. Recall that a bimeromorphic morphism $f\colon X \to Z$ is **small** if $\dim \mathrm{Ex}(f) \le \dim X - 2$, equivalently $f$ is an isomorphism in codimension 1.

**Definition 1.1** (flipping contraction and flip; [?, Def. before Thm. 5.12])**.** Let $(X, B + \boldsymbol{\beta})$ be a $\mathbb{Q}$-factorial generalized log pair. A **flipping contraction** is a small bimeromorphic morphism $f\colon X \to Z$ with $\rho(X/Z) = 1$ such that $-(K_X + B + \boldsymbol{\beta}_X)$ is Kähler over $Z$. The **flip** of $f$ (if it exists) is a small bimeromorphic morphism $f^+\colon X^+ \to Z$ with $\rho(X^+/Z) = 1$ such that $K_{X^+} + B^+ + \boldsymbol{\beta}_{X^+}$ is Kähler over $Z$, where $B^+$ is the strict transform of $B$. The diagram

$$\begin{tikzcd} X \arrow[rr, dashed] \arrow[dr, "f"'] & & X^+ \arrow[dl, "f^+"] \\ & Z & \end{tikzcd}$$

is the **flip diagram**.

**Warning.** Unlike a divisorial contraction, a small contraction cannot be continued: the target of a nontrivial small $(K_X + B)$-negative contraction is never $\mathbb{Q}$-factorial, and a small contraction onto a $\mathbb{Q}$-factorial variety is an isomorphism. This is the structural reason the MMP must be repaired by *replacing* $X$ with the flip $X^+$ rather than passing to $Z$.

**Observation 1.2** (trichotomy of extremal contractions)**.** A birational contraction of a $(K_X + B)$-negative extremal ray is either a divisorial contraction (exceptional locus a divisor) or of **flipping type** (exceptional locus of codimension $\geq 2$). For a compact Kähler 3-fold this dichotomy is read off the null locus of the supporting class $\alpha$: the ray is flipping precisely when $\dim \mathrm{Null}(\alpha) \leq 1$.

## 1.1 Existence of a flip is finite generation

The entire analytic content of "passing from the contraction to the flip" is packaged in the following classical equivalence, which we state and prove in the relative analytic form because it is used identically in both methods below.

**Theorem 1.3** (Kollár–Mori; Birkar)**.** *Let $f : X \to Z$ be a small proper morphism of normal varieties and $D$ a $\mathbb{Q}$-Cartier divisor which is $f$-anti-ample. Then the $D$-flip diagram exists if and only if the relative section algebra*

$$\mathcal{R}(D/Z) = \bigoplus_{m \geq 0} f_* \mathcal{O}_X(\lfloor mD \rfloor)$$

*is a locally finitely generated $\mathcal{O}_Z$-algebra. When it exists the flip is*

$$X^+ = \mathrm{Proj}_Z\, \mathcal{R}(D/Z) \longrightarrow Z,$$

*and it is unique.*

**Idea of proof.** The flip, if it exists, *is* the relative Proj of the section algebra; so the only issue is whether that Proj is of finite type over $Z$ (hence a variety). One direction is the tautology “Proj of a finitely generated algebra”; the reverse direction observes that $X \dashrightarrow X^+$ is an isomorphism in codimension 1, so Hartogs identifies the two section algebras and the flipped divisor $D^+$ is $f^+$-ample — and an ample divisor's section algebra is finitely generated.

**Proof.** Finite generation is local on $Z$, so we may take $Z = \operatorname{Spec} A$ (resp. a Stein base) and $R := \mathcal{R}(D/Z)(Z)$ finitely generated over $A$.
*(⇐)* After truncating we may assume $R$ is generated in degree 1; replacing $D$ by a multiple (which does not change the flip) set $X^+ := \operatorname{Proj} R$ with $f^+\colon X^+ \to Z$ and $\mathcal{O}_{X^+}(1)$ $f^+$-ample. For $m \gg 0$ one recovers $f_*\mathcal{O}_X(mD) = f^+_*\mathcal{O}_{X^+}(m)$, so $X^+$ is normal. Over the smooth locus $V \subset Z$ both $f$ and $f^+$ are isomorphisms (any curve contracted by a flipping contraction lies over the singular locus), so $f^+$ is bimeromorphic. Finally $f^+$ is small: if some divisor $E \subset X^+$ were $f^+$-contracted, then for the divisor $L$ with $\mathcal{O}_{X^+}(1) = \mathcal{O}_{X^+}(L)$ the inclusion $\mathcal{O}_{X^+}(mL) \subsetneq \mathcal{O}_{X^+}(mL+E)$ stays strict after $f^+_*$, because $R^1 f^+_*\mathcal{O}_{X^+}(mL) = 0$ by relative Serre vanishing — contradicting $f^+_*\mathcal{O}_{X^+}(mL) = f_*\mathcal{O}_X(mD)$.
*(⇒)* Given the flip $f^+\colon X^+ \to Z$, the birational map $X \dashrightarrow X^+$ is an isomorphism in codimension 1. On a common open $V \cong V'$ we have $\mathcal{O}(D)|_V \cong \mathcal{O}(D^+)|_{V'}$, and by Hartogs' extension (both spaces are normal, hence $S_2$) $H^0(f^{-1}U, \mathcal{O}(D)) \cong H^0((f^+)^{-1}U, \mathcal{O}(D^+))$ for all $U \subset Z$; thus $\mathcal{R}(D/Z) \cong \mathcal{R}(D^+/Z)$. But $D^+$ is $f^+$-ample, so its relative section algebra is finitely generated. Uniqueness is immediate from $X^+ = \operatorname{Proj}_Z \mathcal{R}(D/Z)$. ■

**Remark 1.4.** In the applications $D = K_X + B + \boldsymbol{\beta}_X$. Theorem **??** reduces the *flip* to local finite generation of the relative log canonical algebra, which is available for Fujiki-class klt spaces (**??**). This is exactly why, in the Kähler world, “the flip is easy and the contraction is hard”.

## 2 The analytic framework and the common engine

The obstruction to importing the projective proofs wholesale is that a nef supporting class of an extremal ray is a transcendental Bott–Chern $(1,1)$-class $\alpha \in H^{1,1}_{\mathrm{BC}}(X)$, not (a multiple of) a $\mathbb{Q}$-divisor, so the base-point-free theorem is unavailable and semi-positivity replaces semi-ampleness. The two methods below are precisely two ways of manufacturing an honest contraction out of such an $\alpha$. Both rest on the same toolbox, which we record now.

**Fact 2.1** (nef supporting class; [?, Prop. 7.3])**.** Let $X$ be a $\mathbb{Q}$-factorial compact Kähler 3-fold with $K_X$ pseudo-effective and $R = \mathbb{R}^+[\Gamma_{i_0}]$ a $K_X$-negative extremal ray of $\overline{\mathrm{NA}}(X)$. Then there is a nef class $\alpha \in N^1(X)$ with $R = \{z \in \overline{\mathrm{NA}}(X) \mid \alpha \cdot z = 0\}$ which is strictly positive on $\big(\overline{\mathrm{NA}}(X)_{K_X \geq 0} + \sum_{i \neq i_0} \mathbb{R}^+[\Gamma_i]\big) \setminus \{0\}$. We call $\alpha$ a **nef supporting class** of $R$.

**Fact 2.2** (Kleiman criterion, Kähler/Fujiki 3-folds)**.** On a compact Kähler (Fujiki-class) 3-fold $X$, a class $\gamma \in H^{1,1}_{\mathrm{BC}}(X)$ is Kähler if and only if $\gamma \cdot z > 0$ for every $z \in \overline{\mathrm{NA}}(X) \setminus \{0\}$.

**Fact 2.3** (null locus = non-Kähler locus; Nakamaye)**.** For a nef and big class $\alpha$ on a compact Kähler manifold, the non-Kähler locus $E_{nK}(\alpha)$ equals the null locus $\bigcup_{\alpha^{\dim Z}\cdot Z=0} Z$. A big class whose non-Kähler locus contains no divisor is **modified Kähler**; it then admits a modification $\mu\colon \widetilde{X} \to X$ with $\mu^*\alpha = \widetilde{\alpha} + E$, where $\widetilde{\alpha}$ is Kähler, $E \geq 0$ is $\mu$-exceptional and $\mu(\operatorname{Supp} E) = E_{nK}(\alpha)$.

**Fact 2.4** (Grauert–Fujiki contractibility)**.** Let $\Xi = \sum m_i D_i$ be a compact analytic divisor in a normal variety $\widehat{X}$ whose conormal bundle $N^*_{\Xi/\widehat{X}} = \mathcal{O}_\Xi(-\Xi)$ is ample. Then there is a bimeromorphic morphism $\psi\colon \widehat{X} \to Y$ contracting $\Xi$ to a point and isomorphic elsewhere. Any contraction is moreover determined by the curves it contracts (rigidity lemma).

**Fact 2.5** (finite generation over Fujiki-class klt bases; [?, Thm. 1.3], [?, Thm. 1.22])**.** Let $f : X \to Y$ be a proper surjective morphism with $X$ in Fujiki's class $\mathcal{C}$ and $(X, B)$ klt. Then $R(X/Y, K_X + B) = \bigoplus_{m\geq 0} f_*\mathcal{O}_X(m(K_X + B))$ is locally finitely generated.

The contraction theorem that organizes the whole discussion is the following; its proof is carried out per contraction type, and the flipping case is the subject of Sections **??**–**??**.

**Theorem 2.6** (contraction theorem for Kähler 3-folds; [?, Thm. 1.2])**.** *Let $(X, B)$ be a $\mathbb{Q}$-factorial compact Kähler 3-fold dlt pair with $K_X + B$ pseudo-effective. Let $\omega$ be a Kähler class such that $\alpha = K_X + B + \omega$ is nef and big and $\alpha^\perp \cap \overline{\mathrm{NA}}(X) = R$ is an extremal ray. Then there is a projective bimeromorphic morphism $f : X \to Z$ with connected fibers such that $\alpha = f^*\alpha_Z$ for a Kähler class $\alpha_Z$ on $Z$. If $f$ is divisorial then $(Z, f_*B)$ is $\mathbb{Q}$-factorial klt; if $R$ is of flipping type then $f$ is a flipping contraction and its flip exists.*

**Theorem 2.7** (existence of the flip; [?, Thm. 5.12])**.** *Let $(X, B + \boldsymbol{\beta})$ be a $\mathbb{Q}$-factorial gdlt pair with $X$ in Fujiki's class $\mathcal{C}$ and $f : X \to Z$ a flipping contraction (**??**). Then the flip $f^+$ of $f$ exists; it is locally projective over $Z$.*

**Proof sketch.** The flipping contraction is (locally) a projective morphism, and $(X, B + \boldsymbol{\beta})$ is klt away from the boundary. By **??** the relative log canonical algebra is locally finitely generated over $Z$; **??** then produces $X^+ = \operatorname{Proj}_Z \mathcal{R}$ locally, and these local flips glue to the global $f^+\colon X^+ \to Z$. ■

*In the Kähler MMP the flip follows from the contraction by finite generation (**??**+**??**); the theorem to prove is the* existence of the flipping contraction.

## 3 Method I — Höring–Peternell (terminal 3-folds)

Throughout this section $X$ is a *terminal* compact Kähler 3-fold with $K_X$ pseudo-effective, $R = \mathbb{R}^+[\Gamma_{i_0}]$ a small $K_X$-negative extremal ray, and $\alpha$ its nef supporting class (**??**). Being terminal, $X$ has only isolated (rational, Gorenstein-in-codimension-2) singularities, so the geometry localizes at the finitely many curves in $R$.

**Idea of proof.** The strategy is entirely analytic and avoids any auxiliary MMP. First show the supporting class $\alpha$ is nef and big with 1-dimensional non-Kähler locus $C$ (the curve swept by $R$); the key numerical input is that $\alpha^2 \cdot S > 0$ for every surface $S$ (**??**), so no surface can be $\alpha$-trivial. Then $\alpha$ is modified Kähler, so a modification splits $\mu^*\alpha = \widetilde{\alpha} + E$; on the preimage of each component of $C$ the class $-E$ restricts to the Kähler class $\widetilde{\alpha}$, giving an *ample conormal bundle*. Grauert's contractibility theorem then contracts $C$, and the rigidity lemma descends the contraction to $X$.

**Proposition 3.1** (positivity on surfaces; [?, Prop. 7.11])**.** *Suppose $R$ is small with nef supporting class $\alpha$, and let $S \subset X$ be an irreducible surface. Then $\alpha^2 \cdot S > 0$.*

**Proof.** Since $\alpha$ is nef, $\alpha^2 \cdot S \geq 0$; suppose for contradiction $\alpha^2 \cdot S = 0$. By **??** the class $\alpha - K_X$ is positive on $\overline{\mathrm{NA}}(X) \setminus \{0\}$ after rescaling, hence Kähler by **??**; write $\omega := \alpha - K_X$. If $\alpha|_S = 0$ then $-K_X|_S = \omega|_S$ is ample, so $S$ is projective and covered by curves whose classes lie in $R$ (as $\alpha|_S = 0$), contradicting smallness of $R$.
So assume $\alpha|_S \neq 0$. From $0 = \alpha^2 \cdot S = K_X \cdot \alpha \cdot S + \omega \cdot \alpha \cdot S$ and $\omega \cdot \alpha \cdot S = \omega|_S \cdot \alpha|_S > 0$ (Hodge index) we get $K_X \cdot \alpha \cdot S < 0$. Let $\pi \colon S' \to S$ be the composition of normalization and minimal resolution, so $K_{S'} + E = \pi^* K_S$ with $E \geq 0$; then $K_{S'} \cdot \pi^*\alpha|_S \leq K_S \cdot \alpha|_S < 0$. Hence $K_{S'}$ is not pseudo-effective, $S'$ is uniruled with $H^2(S', \mathcal{O}_{S'}) = 0$, and $\pi^*\alpha$ is a genuine nef $\mathbb{R}$-divisor on the projective surface $S'$, an element of $\overline{\mathrm{NM}}(S')$ positive on every movable curve (the ray $R$ carries only finitely many curve classes). Fixing an ample $A$ and applying the BDPP decomposition $\pi^*\alpha = C_\varepsilon + \sum \lambda_{i,\varepsilon} M_{i,\varepsilon}$ with $(K_{S'} + \varepsilon A) \cdot C_\varepsilon \geq 0$ and $M_{i,\varepsilon}$ movable: since $(\pi^*\alpha)^2 = 0$ and $\pi^*\alpha \cdot M_{i,\varepsilon} > 0$ we must have $\pi^*\alpha = C_\varepsilon$ for all $\varepsilon$, whence $K_{S'} \cdot \pi^*\alpha \geq 0$ in the limit — contradicting $K_{S'} \cdot \pi^*\alpha < 0$. ■

**Theorem 3.2** (the flipping contraction; [?, Thm. 7.12])**.** *Suppose $R$ is small and let $C$ be the locus covered by the curves in $R$. Then $C$ is contractible: there is a flipping contraction $\varphi \colon X \to Y$ with $\mathrm{Ex}(\varphi) = C$. Consequently, by **??**, the flip of $\varphi$ exists.*

**Proof.** *$\alpha$ is nef and big.* By **??**, $-K_X + \alpha$ is positive on $\overline{\mathrm{NA}}(X) \setminus \{0\}$ after rescaling, hence Kähler (**??**). Thus $\alpha = K_X + (-K_X + \alpha)$ is the sum of a pseudo-effective and a Kähler class, so big; combined with nefness, $\alpha^3 > 0$.
*The non-Kähler locus is the curve $C$.* Let $\pi\colon \widehat{X} \to X$ be a desingularization. By **??**, $E_{nK}(\pi^*\alpha)$ is the null locus of the nef and big class $\pi^*\alpha$. If $Z \subset \widehat{X}$ is a surface with $(\pi^*\alpha)^2 \cdot Z = 0$ then, by the projection formula and **??**, $\dim \pi(Z) \le 1$; hence $E_{nK}(\pi^*\alpha)$ is a finite union of $\pi$-exceptional surfaces and curves. Since $E_{nK}(\alpha) \subset \pi(E_{nK}(\pi^*\alpha))$ and $X_{\mathrm{sing}}$ is finite (terminal $\Rightarrow$ isolated), we conclude $E_{nK}(\alpha) = \bigcup_{\alpha \cdot B = 0} B = C$, a curve.
*$\alpha$ is modified Kähler; produce an ample conormal bundle.* As the big class $\alpha$ has non-Kähler locus containing no divisor, it is modified Kähler (**??**): there is $\mu\colon \widetilde{X} \to X$ and a Kähler $\widetilde{\alpha}$ with $\mu^*\alpha = \widetilde{\alpha} + E$, $E \ge 0$ $\mu$-exceptional and $\mu(\operatorname{Supp} E) = C$. For a connected component $Z \subset C$ put $D_Z := \operatorname{Supp} \mu^{-1}(Z)$. Since $\alpha|_Z \equiv 0$ we get $-E|_{D_Z} = \widetilde{\alpha}|_{D_Z}$, which is ample. Perturbing the coefficients of $E$ (ampleness is open) and clearing denominators, we obtain an integral $E' = \sum m_i D_i + R$ with $D_i \subset D_Z$, $R$ disjoint from $D_Z$, and $-E'|_{D_Z}$ ample. Hence the conormal bundle $N^*_{\sum m_i D_i / \widehat{X}} = \mathcal{O}(-\sum m_i D_i)$ is ample (ampleness may be tested on the reduction).
*Contract.* By Grauert–Fujiki contractibility (**??**) there is $\psi\colon \widehat{X} \to Y_Z$ contracting $\sum m_i D_i$ to a point; by the rigidity lemma there is $\varphi_Z\colon X \to Y_Z$ with $\psi = \varphi_Z \circ \mu$. Running this over all components of $C$ yields the desired contraction $\varphi\colon X \to Y$ with $\mathrm{Ex}(\varphi) = C$. As $\varphi$ is small and $K_X$-negative on $R$, it is a flipping contraction, and its flip exists by **??**. ■

# 4 Method II — Das–Hacon (klt / dlt log pairs)

Das and Hacon handle log pairs $(X, B)$, where boundary components and non-isolated centres appear and the direct positivity argument of Section **??** is not available. Their device is to *bootstrap the log MMP itself*: pass to a dlt modification and run an auxiliary relative MMP steered by nef thresholds on the boundary surfaces, then descend. We first record the short terminal route, then the substantive klt route.

## 4.1 The terminal route

**Fact 4.1** (contraction from a big nef class with 1-dimensional null locus; [?, Prop. 4.5]). Let $X$ be a normal compact Kähler variety with isolated rational singularities and $\alpha$ a nef and big $(1,1)$-class. If $\mathrm{Null}(\alpha)$ is a pure 1-dimensional analytic subset, then there is a proper bimeromorphic morphism $f\colon X \to Y$ with $\mathrm{Ex}(f) = \mathrm{Null}(\alpha)$ contracting $\mathrm{Null}(\alpha)$ to finitely many points.

**Corollary 4.2** (terminal flipping contraction; [?, Cor. 4.6]). *Let $(X, B)$ be a compact Kähler 3-fold terminal pair, $\omega$ a Kähler class and $\alpha = K_X + B + \omega$ nef but not Kähler with* $\mathrm{Null}(\alpha)$

*pure* 1*-dimensional and* $\overline{\mathrm{NA}}(X) \cap \alpha^{\perp} = R$ *an extremal ray. Then the flipping contraction* $f: X \to Z$ *and its flip* $f^{+}: X^{+} \to Z$ *both exist.*

**Proof.** A terminal 3-fold has isolated rational singularities, and $\alpha$ is big because its null locus is 1-dimensional (so the non-Kähler locus contains no divisor). By **??** there is a small contraction $f: X \to Z$ with $\mathrm{Ex}(f) = \mathrm{Null}(\alpha)$; being $(K_X + B)$-negative and small, it is a flipping contraction. The flip is then obtained from **??**+**??**: construct the local log canonical models $\mathrm{Proj}_Z \mathcal{R}$ over $Z$ and glue them to $f^{+}$. ■

## 4.2 The klt/dlt route

**Fact 4.3** (plt contraction theorem; [**?**, Thm. 1.1])**.** Let $(X, S + B)$ be a $\mathbb{Q}$-factorial compact Kähler plt pair and $\pi: S \to T$ a contraction such that $-(K_X + S + B)|_S$ and $-S|_S$ are $\pi$-ample. Then there is a projective bimeromorphic morphism $p: X \to Z$ with connected fibers such that $p|_S = \pi$ and $p|_{X \setminus S}$ is an isomorphism; if $p$ is of flipping type, its flip exists.

This is the analytic engine that turns a contraction *on a boundary surface* into a contraction of the ambient 3-fold. It is available in every dimension and is what powers the induction below.

**Theorem 4.4** (klt flipping contraction; [**?**, Thm. 6.2])**.** *Let* $(X, B)$ *be a strongly* $\mathbb{Q}$*-factorial compact Kähler* 3*-fold klt pair with* $K_X + B$ *pseudo-effective. Suppose* $\alpha = [K_X + B + \beta]$ *is nef and big for some Kähler* $\beta$ *and* $\alpha^{\perp} \cap \overline{\mathrm{NA}}(X) = R$ *is an extremal ray of flipping type (i.e.* $\dim \mathrm{Null}(\alpha) \leq 1$*). Then the flipping contraction* $f: X \to Z$ *and the flip* $X \dashrightarrow X^{+}$ *exist, and there is a Kähler class* $\alpha_Z$ *on* $Z$ *with* $\alpha = f^{*}\alpha_Z$*.*

**Idea of proof.** Resolve to a dlt modification $\nu: X' \to X$ and detect, on each boundary surface $S' \subset \lfloor \Delta' \rfloor$, a negative extremal ray via a positive *nef threshold* $\tau_{S'}$ (surface cone theorem + adjunction). Extending the resulting surface contraction to $X'$ by the plt contraction theorem (**??**) gives one step of an auxiliary MMP; running it produces a bimeromorphic $\phi: X' \dashrightarrow X^{+}$. The point is to control this MMP so that it is *proper over* $U = X \setminus \mathrm{Null}(\alpha)$ and contracts *exactly* the $\nu$-exceptional divisors — then it descends to a small map $\psi: X \dashrightarrow X^{+}$. A final Grauert contraction of a common-resolution difference class builds $X^{+} \to Z$ and the induced $f, f^{+}$, after which one checks the numerical properties.

**Architecture of the proof.** *Step 0 (dlt modification).* Take $\nu: X' \to X$ a $\mathbb{Q}$-factorial dlt modification with $\nu^{*}\alpha = \omega' + G'$, $G' \geq 0$ $\nu$-exceptional and $\omega'$ a modified Kähler class, arranged so that $(X', \Delta' + t\omega')$ is gdlt and $K_{X'} + \Delta' + t\omega' + a\alpha'$ is nef for fixed $t, a > 0$, with $\operatorname{Supp} G' = \operatorname{Ex}(\nu) = \lfloor \Delta' \rfloor$.

*Step 1 (find the surface contraction).* For a component $S' \subset \lfloor \Delta' \rfloor$, adjunction gives $K_{S'} + \Delta_{S'} + t\omega'_{S'} = (K_{X'} + \Delta' + t\omega')|_{S'}$ with $(S', \Delta_{S'})$ dlt and $\omega'_{S'}$ big. Define the nef threshold

$$\tau_{S'} := \inf\{s \geq 0 \mid K_{S'} + \Delta_{S'} + a\alpha_{S'} + s\omega_{S'} \text{ is nef for some } a > 0\}.$$

By the cone theorem for generalized Kähler DLT surfaces, if $\tau_{S'} > 0$ there is a $(K_{S'} + \Delta_{S'})$-negative extremal ray $R_{S'}$ with $(K_{S'} + \Delta_{S'} + \tau_{S'}\omega_{S'}) \cdot R_{S'} = \alpha_{S'} \cdot R_{S'} = 0$. Put $\tau = \max_{S'} \tau_{S'}$; if $\tau > 0$ then $K_{X'} + \Delta' + \tau\omega' + a\alpha'$ is nef for some $a > 0$ and a component $S'$ carries an extremal ray $R_i$ with $(K_{X'} + \Delta' + \tau\omega') \cdot R_i = \alpha' \cdot R_i = 0$.

*Step 2 (MMP step).* By the plt contraction theorem (**??**) the surface contraction $\pi \colon S' \to T$ extends to a bimeromorphic $p \colon X' \to Z'$ with $p|_{S'} = \pi$ and $p|_{X' \setminus S'}$ an isomorphism; $p$ is divisorial or flipping, and in the flipping case its flip exists. Iterating yields a $(K_{X'} + \Delta' + t\omega' + a\alpha')$-MMP $\phi \colon X' \dashrightarrow X^+$ (for $0 < t \leq \varepsilon$), terminating when $\tau = 0$.

*Step 3 (properness over $U$ and descent).* With $U = X \setminus \operatorname{Null}(\alpha)$ and $X'_U = \nu^{-1}(U)$, every step is vertical over $U$: $\phi_U = \phi|_{X'_U}$ is a proper bimeromorphic map over $U$. Moreover the divisors contracted by $\phi$ coincide with the $\nu$-exceptional divisors, so the induced $\psi \colon X \dashrightarrow X^+$ is a *small* bimeromorphic map of strongly $\mathbb{Q}$-factorial varieties, isomorphic outside $\operatorname{Null}(\alpha)$.

*Step 4 (build the contraction).* On a common resolution $p \colon W \to X$, $q \colon W \to X^+$ form $E = p^*\eta - q^*\eta^+$; its positivity lets Grauert's theorem contract it, producing $g \colon W \to Z$ and hence induced morphisms $f \colon X \to Z$ and $f^+ \colon X^+ \to Z$.

*Step 5 (numerical properties).* By construction $f$ contracts a nonempty set of $\alpha$-trivial curves; one checks it contracts *every* curve in $R$, is $(K_X + B)$-negative, and satisfies $\alpha = f^*\alpha_Z$ (base-point-freeness) with $\alpha_Z$ Kähler.

*Step 6 (Kählerness after the flip).* Since $\psi_*$ is an isomorphism, $\rho(X^+/Z) = 1$; $f^+$ contracts a $(K_{X^+} + B^+)$-positive curve, so $K_{X^+} + B^+$ is $f^+$-ample, and $(f^+)^*\alpha_Z + \delta(K_{X^+} + \Delta^+)$ is Kähler for $0 < \delta \ll 1$; hence $X^+$ is again a Kähler 3-fold. ■

**Remark 4.5** (endpoint)**.** The clean end statement is **??**: any $\mathbb{Q}$-factorial gdlt flipping contraction of a Fujiki-class variety admits its flip, locally projective over $Z$. The generalized-pair and relative refinements for Kähler 3-folds are due to Das–Hacon–Yáñez [**?**], who additionally bound the flipping curve by $0 > (K_X + B + \boldsymbol{\beta}_X) \cdot C \geq -6$.

## 5 Comparison of the two methods

**Observation 5.1** (Höring–Peternell vs. Das–Hacon)**.** The two methods produce the *same* object — the flipping contraction — by opposite philosophies.

- **Höring–Peternell** is a *direct analytic contraction*. One proves the supporting class is modified Kähler with a 1-dimensional non-Kähler locus (**??**), extracts an ample conormal bundle, and contracts by Grauert (**??**). It is clean and geometric but is tied to the terminal (non-log) case, where singularities are isolated and the null locus is automatically a curve.

- **Das–Hacon** is an *induction that bootstraps the log MMP*. One passes to a dlt modification, runs an auxiliary MMP steered by nef thresholds on boundary surfaces (surface cone theorem + plt contraction theorem, **??**), forces properness away from the null locus, and descends (**??**). It is heavier machinery but reaches klt/dlt log pairs and generalized pairs, where boundary components and non-isolated centres occur.

**Key takeaways**

- In the Kähler MMP the *flip* is soft: given a flipping contraction it is $\mathrm{Proj}_Z$ of a locally finitely generated relative log canonical algebra (**??**, **??**). The work is the *flipping contraction*.
- Both methods share the analytic core: reduce the flip to local finite generation, and realize contractions via *Grauert–Fujiki contractibility + the rigidity lemma*.
- They diverge only in how the flipping contraction is produced — Höring–Peternell contract the curve *directly* (ample conormal bundle), Das–Hacon produce it *indirectly* by an auxiliary MMP on a dlt modification.
- Reality check: flips are genuinely a *singular / log* 3-fold phenomenon — for smooth projective 3-folds there are no flips; the first honest flips appear on singular or log 3-folds and (e.g. toric) smooth 4-folds.